

ZAYAN AHMAD GHOUS

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GitHub: github.com/ZayanAhmadGhous | Portfolio: zayanahmad.me

PROFESSIONAL SUMMARY

BSCS student with hands-on experience in Machine Learning, Data Engineering, and MLOps. Skilled in building machine learning model, deploying machine learning models, and developing REST APIs using Python, TensorFlow, and FastAPI. Experienced with cloud-based ML applications and focused on delivering scalable, efficient solutions. Seeking an opportunity to apply technical expertise while gaining industry experience.

TECHNICAL SKILLS

Programming: Python, SQL, C++, C
AI & Machine Learning: Machine Learning, Deep Learning, TensorFlow, Keras, Scikit-learn, CNNs, Classification, Regression, Model Optimization, Cross-Validation
Data & ML Engineering: NumPy, Pandas, Data Preprocessing, Feature Engineering, EDA, Data Cleaning, Feature Scaling, Categorical Encoding, Data Augmentation
Backend & APIs: FastAPI, Django REST Framework, REST API Design, HTTP Methods, Postman, Gradio
Deployment & Tools: Docker, Git, GitHub, Linux, Hugging Face Spaces, Streamlit, Jupyter Notebook, Google Colab
Data Visualization: Matplotlib

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern
University of Engineering & Technology (UET), Lahore May 2025 – July 2025

- Developed and trained machine learning models including Decision Trees, Random Forests, and Neural Networks using Python and scikit-learn, achieving 85%+ accuracy on real-world datasets.
- Built end-to-end data preprocessing pipelines for datasets with 10,000+ samples, including missing-value handling, feature scaling, categorical encoding, and data transformation.
- Developed CNN-based image classification models for MRI brain scan analysis using TensorFlow/Keras and data augmentation techniques.
- Deployed ML applications using FastAPI and Gradio, enabling real-time model inference through REST APIs and interactive interfaces.

Machine Learning Engineer - Freelance Collaboration July 2025 –December 2025

- Developed classification and regression solutions using Python and scikit-learn based on client requirements.
- Performed data preprocessing, feature engineering, and exploratory data analysis to prepare datasets for model development.
- Built REST APIs with FastAPI to integrate machine learning models with application interfaces.
- Containerized ML applications using Docker and collaborated with clients to gather requirements and deliver AI solutions.

PROJECTS

Brain Tumor Detection - CNN + FastAPI + Deployment **Live:** huggingface.co/spaces/Ghous/brain_tumor

- Built a CNN-based brain tumor classification model achieving 88% accuracy on MRI dataset (500+ images) using TensorFlow/Keras.
- Improved model generalization using data augmentation, dropout regularization, batch normalization, and early stopping.
- Developed REST API using FastAPI for real-time inference and integrated with a Gradio UI for user-friendly predictions.
- Deployed end-to-end ML application on Hugging Face Spaces for public web access.

House Price Prediction API - Machine Learning + FastAPI + Docker **Docker Hub:** [zayanahmad123/house-price-api](https://hub.docker.com/u/zayanahmad123)

- Built a house price prediction model using sk-learn with feature engineering and categorical encoding on a real-world housing dataset.
- Developed automated preprocessing pipelines using Pandas, SciPy, and MEstimateEncoder for reliable model training and inference.
- Created a FastAPI-based REST API for real-time house price predictions with production-ready inference workflows.
- Containerized the complete application using Docker and published the image on Docker Hub for reproducible deployment.

EDUCATION

Board of Intermediate and Secondary Education **Lahore, Pakistan**
Intermediate in Computer Sciences (Punjab Group of colleges) 2022 – 2024

Minhaj University Lahore **Lahore, Pakistan**
BS in Computer Sciences 2024 – Present